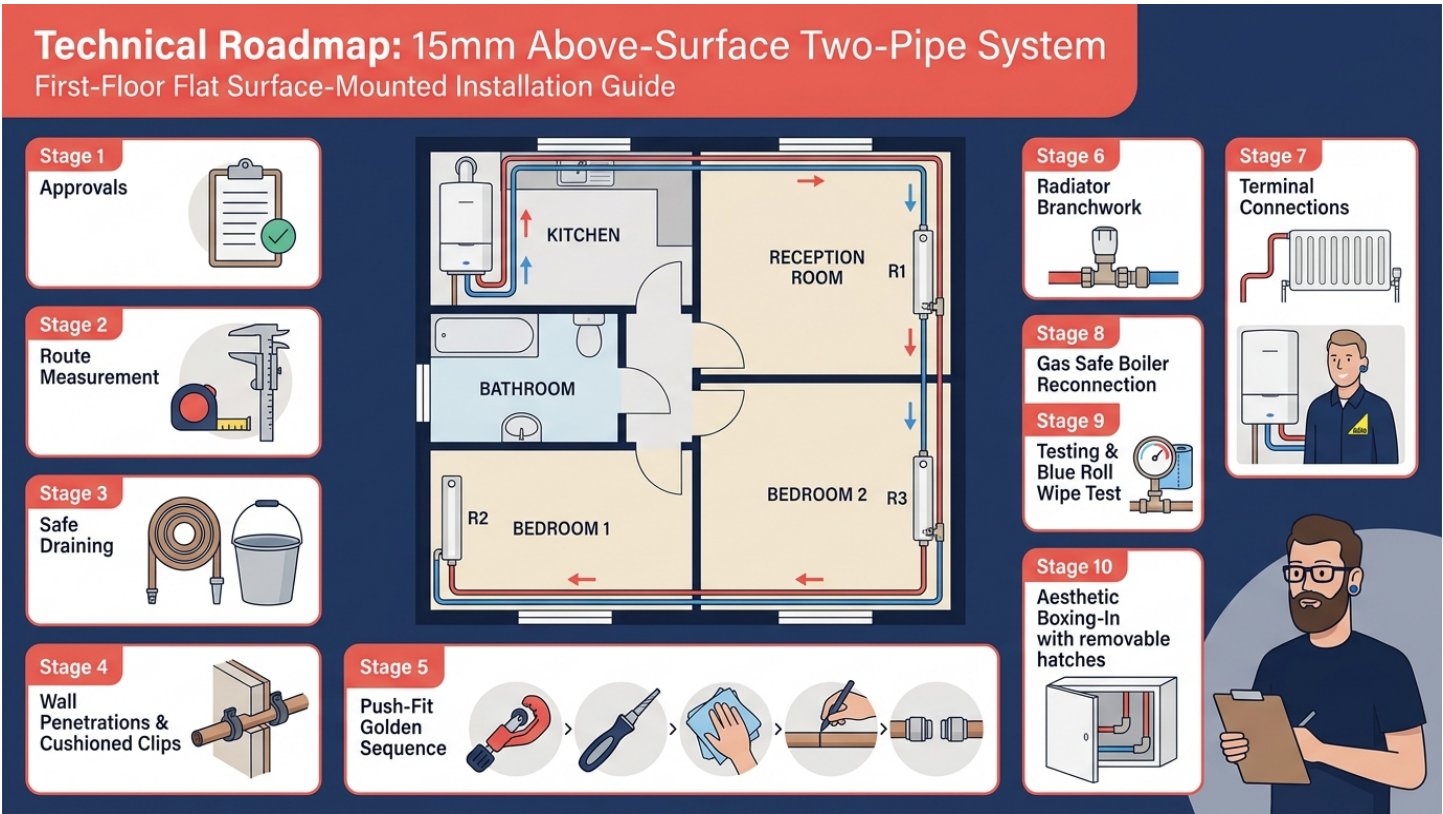


Central Heating Water Reroute Roadmap

Technical Installation Manual · Above-Surface 15mm Copper Parallel Two-Pipe System (V5)





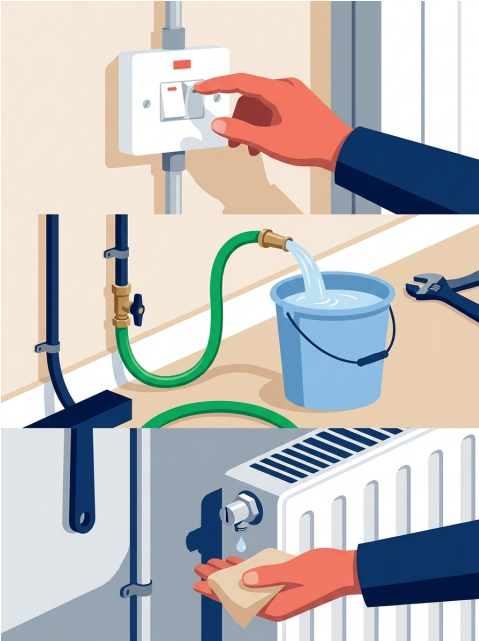
- Secure written consent from freeholder/managing agent regarding above-surface pipework.
- Notify home insurer in writing to ensure building coverage remains active.
- Download and print Glow-worm Compact 24c installation manual.
- Obtain professional plumber verification of 15mm route before ordering materials.

Stage 2: Route Measurement & Precision Planning



- Physically measure perimeter wall runs at skirting level with a tape measure.
- Double the one-way distance (~22.8m) to calculate total pipe needed (~45.6m).
- Add 10% waste allowance and purchase 17 x 3m lengths (51m total).
- Verify existing boiler stubs with vernier calipers (15mm OD).

Stage 3: Safe Draining & Isolation



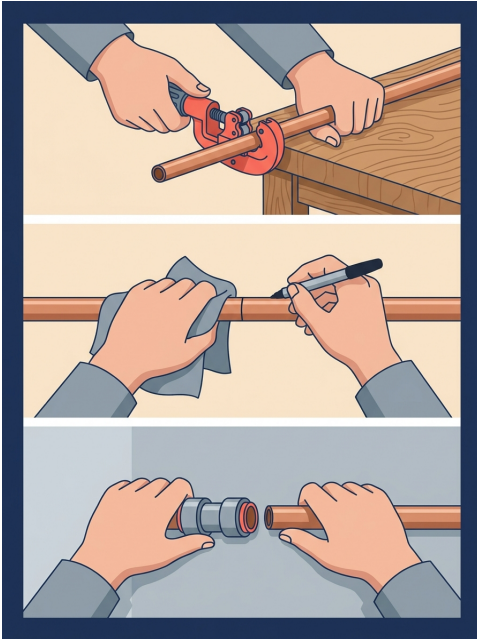
- Switch off boiler at programmer and fused spur switch (confirm 0V with voltage tester).
- Allow system to cool for at least 1 hour to prevent scalding.
- Attach hose to lowest drain-off valve, open bleed valves, and drain completely until 0 bar.

Stage 4: Route & Wall Penetrations



- Scan both faces of partition walls using a Bosch Truvo detector to locate concealed cables.
- Drill wall penetrations at skirting board level and install protective grommets/sleeves.
- Mount rubber-lined cushioned pipe clips at 600mm horizontal and 1200mm vertical intervals.

Stage 5: Push-Fit Golden Sequence



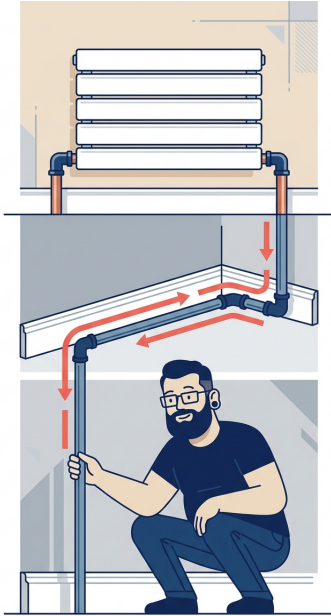
- Step 1: Cut square using Rothenberger Pipeslice (never use a hacksaw).
- Step 2: Deburr internal and external copper edges with a reamer.
- Step 3: Clean & inspect tube end with a dry cloth (scratch-free mirror finish).
- Step 4: Mark insertion depth (16mm for Sprint / 23mm for Classic) using permanent marker.
- Step 5: Push firmly into fitting until depth mark meets socket collar, verify with pull-check.

Stage 6: Radiator Branchwork



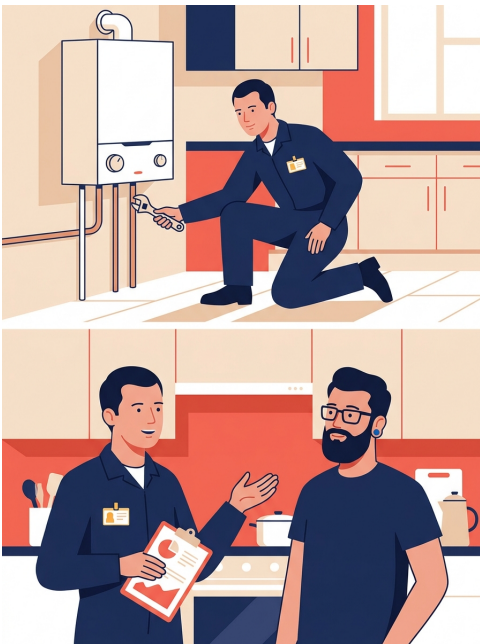
- Splice Tectite Sprint Equal Tees into the parallel 15mm flow and return main lines.
- Run short copper branch pipes vertically up from tees to radiator valves.
- Connect branches to TRV and lockshield valves using compression nuts and two spanners.

Stage 7: R3 Terminal Connection & Return Trace



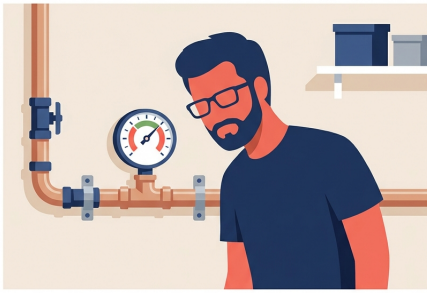
- Install terminal 90° elbows at Radiator R3 (Bedroom 2) — no continuation past R3.
- Trace return line continuously along Bedroom 2, Bedroom 1, Reception Room back to boiler.
- Document all visible pipework and joints before proceeding to boiler connection.

Stage 8: Boiler Connections (Gas Safe Boundary)



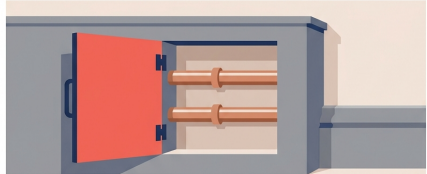
- Stop at the qualified-work boundary: Gas Safe engineer must make or supervise boiler connections.
- Clean-shaven Gas Safe engineer identifies flow and return stubs under boiler using manual.
- Record engineer credentials, Gas Safe registration number, and site sign-off.

Stage 9: Testing, Refill & Monitoring



- Refill system via boiler filling loop to 1.0–1.5 bar under engineer supervision.
- Perform the Blue Roll Wipe Test: wipe dry tissue around every joint to detect moisture.
- Dose system with approved inhibitor and observe pressure stability during hot/cold cycles.

Stage 10: Aesthetic Boxing-In & Access



- Construct 160mm wide timber batten framework along skirting level.
- Screw 6mm MDF cover panels to battens, leaving 5–10mm thermal expansion gaps.
- Install magnetic removable access hatches (min 150x150mm) over all fittings and valves.